

Symmetries and Conservation HW-5

PROBLEM 1. Calculate the matrix $e^{i\mathbf{r}\cdot\boldsymbol{\sigma}}$, where $\boldsymbol{\sigma}$ are three Pauli matrices,

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3) \quad \text{and} \quad \mathbf{r} = (x, y, z).$$

SOLUTION:

$$e^{i\mathbf{r}\cdot\boldsymbol{\sigma}} = e^i \begin{bmatrix} z & x - yi \\ x + yi & -z \end{bmatrix}$$

Diagonalizing the matrix:

$$\lambda^2 - z^2 - x^2 - y^2 = 0 \implies \lambda = \pm r$$

eigenvectors:

$$\lambda = r : v = \left(\frac{x - yi}{\sqrt{2r^2 - 2rz}}, \frac{r - z}{\sqrt{2r^2 - 2rz}} \right), \lambda = -r : v = \left(\frac{-x + yi}{\sqrt{2r^2 + 2rz}}, \frac{r + z}{\sqrt{2r^2 + 2rz}} \right)$$

$$i \begin{bmatrix} z & x - yi \\ x + yi & -z \end{bmatrix} = \begin{bmatrix} \frac{x-yi}{\sqrt{2r^2-2rz}} & \frac{-x+yi}{\sqrt{2r^2+2rz}} \\ \frac{r-z}{\sqrt{2r^2-2rz}} & \frac{r+z}{\sqrt{2r^2+2rz}} \end{bmatrix} \begin{bmatrix} ir & 0 \\ 0 & -ir \end{bmatrix} \begin{bmatrix} \frac{x+yi}{\sqrt{2r^2-2rz}} & \frac{r-z}{\sqrt{2r^2-2rz}} \\ \frac{-x-yi}{\sqrt{2r^2+2rz}} & \frac{r+z}{\sqrt{2r^2+2rz}} \end{bmatrix} = UDU^\dagger$$

Thus

$$\begin{aligned} e^{i\mathbf{r}\cdot\boldsymbol{\sigma}} &= Ue^DU^\dagger = \begin{bmatrix} \frac{x-yi}{\sqrt{2r^2-2rz}} & \frac{-x+yi}{\sqrt{2r^2+2rz}} \\ \frac{r-z}{\sqrt{2r^2-2rz}} & \frac{r+z}{\sqrt{2r^2+2rz}} \end{bmatrix} \begin{bmatrix} e^{ir} & 0 \\ 0 & e^{-ir} \end{bmatrix} \begin{bmatrix} \frac{x+yi}{\sqrt{2r^2-2rz}} & \frac{r-z}{\sqrt{2r^2-2rz}} \\ \frac{-x-yi}{\sqrt{2r^2+2rz}} & \frac{r+z}{\sqrt{2r^2+2rz}} \end{bmatrix} \\ &= \begin{bmatrix} \cos r + i \sin r \frac{z}{r} & \frac{x+yi}{r} i \sin r \\ \frac{x-yi}{r} i \sin r & \cos r - i \sin r \frac{z}{r} \end{bmatrix} \end{aligned}$$

where $r = \sqrt{x^2 + y^2 + z^2}$.

PROBLEM 2. $SU(2)$ algebra: Show explicitly that the spin 1 representation ($|1, m\rangle$, where $m = 0, \pm 1$) obtained by the highest weight procedure with $j = 1$ is equivalent to the adjoint representation, with $f_{abc} = \epsilon_{abc}$ by finding the similarity transformation that implements the equivalence.

SOLUTION:

Consider the following isomorphism (linear transformation) of the representation space, it suffices to give the transformation of the basis:

$$\begin{aligned} |1, 1\rangle &\rightarrow |J_+\rangle = \frac{1}{\sqrt{2}}(J_1 + iJ_2)\rangle \\ |1, 0\rangle &\rightarrow |J_3\rangle \\ |1, -1\rangle &\rightarrow |J_-\rangle = -\frac{1}{\sqrt{2}}(J_1 - iJ_2)\rangle \end{aligned}$$

We show that under each basis, the representation matrix of J_1, J_2, J_3 are the same.

$$\begin{aligned}
J_3 &= |1, 1\rangle \langle 1, 1| - |1, -1\rangle \langle 1, -1|, ad_{J_3} = |[J_3, J_{3,-,+}] \rangle \langle J_{3,-,+}| = |J_+\rangle \langle J_+| - |J_-\rangle \langle J_-| \\
J_1 &= \frac{1}{\sqrt{2}}(|1, 1\rangle \langle 1, 0| + |1, 0\rangle \langle 1, -1| + h.c.), ad_{J_1} = |[J_1, J_{3,-,+}] \rangle \langle J_{3,-,+}| \\
&= \frac{1}{\sqrt{2}}(|J_- + J_+\rangle \langle J_3| + |J_3\rangle \langle J_-| + |J_3\rangle \langle J_+|) \\
J_2 &= \frac{1}{\sqrt{2}i}(|1, 1\rangle \langle 1, 0| + |1, 0\rangle \langle 1, -1| - h.c.), ad_{J_2} = |[J_2, J_{3,-,+}] \rangle \langle J_{3,-,+}| \\
&= \frac{1}{\sqrt{2}i}(|-J_- + J_+\rangle \langle J_3| + |J_3\rangle \langle J_-| - |J_3\rangle \langle J_+|)
\end{aligned}$$

PROBLEM 3. Suppose that $(\sigma_a)_{ij}$ and $(\eta_a)_{ij}$ are Pauli matrices in two different 2-D spaces. In the four dimensional tensor product space, define the basis

$$\begin{aligned}
|1\rangle &= |i=1\rangle|x=1\rangle \\
|2\rangle &= |i=1\rangle|x=2\rangle \\
|3\rangle &= |i=2\rangle|x=1\rangle \\
|4\rangle &= |i=2\rangle|x=2\rangle
\end{aligned}$$

Write out the matrix elements of $\sigma_2 \otimes \eta_1$ in this basis.

SOLUTION:

$$\begin{aligned}
\sigma_2 |i=1\rangle &= i |i=2\rangle, \sigma_2 |i=2\rangle = -i |i=1\rangle \\
\eta_1 |x=1\rangle &= |x=2\rangle, \eta_1 |x=2\rangle = |x=1\rangle
\end{aligned}$$

Thus

$$\sigma_2 \otimes \eta_1 |1\rangle = i |4\rangle, \sigma_2 \otimes \eta_1 |2\rangle = i |3\rangle, \sigma_2 \otimes \eta_1 |3\rangle = -i |2\rangle, \sigma_2 \otimes \eta_1 |4\rangle = -i |1\rangle$$

and

$$\sigma_2 \otimes \eta_1 = \begin{bmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{bmatrix}$$

PROBLEM 4. We will often abbreviate the tensor product notation by leaving out the indices and the identity matrices. This makes for a very compact notation, but you must keep your wits about you to stay in the right space. In the example of previous problem, we could write $(\sigma_a)_{ij} (\eta_b)_{xy}$ as $\sigma_a \eta_b$, $(\sigma_a)_{ij} \delta_{xy}$ as σ_a , $\delta_{ij} (\eta_b)_{xy}$ as η_b , and $\delta_{ij} \delta_{xy}$ as 1. So for example, $(\sigma_1) (\sigma_2 \eta_1) = i \sigma_3 \eta_1$ and $(\sigma_1 \eta_2) (\sigma_1 \eta_3) = i \eta_1$. Calculate

$$\begin{aligned}
&[\sigma_a, \sigma_b \eta_c] \\
&\text{Tr}(\sigma_a \{\eta_b, \sigma_c \eta_d\}) \\
&[\sigma_1 \eta_1, \sigma_2 \eta_2]
\end{aligned}$$

where $\{A, B\} = AB + BA$ is the anti-commutator.

SOLUTION:

We should notice that $AB \otimes C = BA \otimes C + [A, B] \otimes C$, thus (because here δ commute with η_c)

$$[\sigma_a, \sigma_b \eta_c] = [\sigma_a, \sigma_b] \otimes \eta_c = 2i\epsilon_{abd}\sigma_d \otimes \eta_c$$

$$\sigma_a \{\eta_b, \sigma_c \eta_d\} = \sigma_a \sigma_c \otimes \{\eta_b, \eta_d\} = \sigma_a \sigma_c \otimes 2\delta_{bd} \implies \text{Tr}(\sigma_a \{\eta_b, \sigma_c \eta_d\}) = 2\delta_{bd} \text{Tr}(\sigma_a \sigma_c) = 4\delta_{bd}\delta_{ac}$$

$$[\sigma_1 \eta_1, \sigma_2 \eta_2] = \sigma_1 \sigma_2 \otimes \eta_1 \eta_2 - \sigma_2 \sigma_1 \otimes \eta_2 \eta_1 = -\sigma_3 \otimes \eta_3 + \sigma_3 \otimes \eta_3 = 0$$

PROBLEM 5. In class, we have write down the 8 generators for $SU(3)$ group. If you missed the class, or forget them, search "Gell-Mann matrices". Note that the generators are defined as $T_a = \frac{1}{2}\lambda_a$. Explicitly calculate the structural constant f_{147} and f_{458} .

SOLUTION:

$$[T_1, T_4] = \frac{1}{4} \left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = \frac{1}{2} iT_7 \implies f_{147} = \frac{1}{2}$$

$$[T_4, T_5] = \frac{1}{4} \left(\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right) = \frac{1}{4} \begin{bmatrix} 2i & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2i \end{bmatrix} = \frac{i}{2} (T_3 + \sqrt{3}T_8) \\ \implies f_{458} = \frac{\sqrt{3}}{2}$$

PROBLEM 6. Consider the $SU(2)$ subalgebra of $SU(3)$.

- (10 points) Show that T_1, T_2 , and T_3 generate and $SU(2)$ subalgebra of $SU(3)$.

- (20 points) Every representation of $SU(3)$ must also be a representation of the subalgebra. However, the irreducible representations of $SU(3)$ are not necessarily irreducible under the subalgebra. How does the representation generated by the Gell-Mann matrices transform under this subalgebra? That is, reduce, if necessary, the three dimensional representation into representations which are irreducible under the subalgebra and state which irreducible representations appear in the reduction.

- (20 points) Answer the same question for the adjoint representation of $SU(3)$. That is, How does the the adjoint representation of $SU(3)$ transform under this subalgebra?

SOLUTION:

It suffices to prove that T_1, T_2, T_3 forms a closed algebra and satisfies the relation

$$[T_a, T_b] = i\epsilon_{abc}T_c$$

By direct calculation we see that

$$[T_1, T_2] = \frac{1}{4} \left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} i & 0 & 0 \\ 0 & -i & 0 \\ 0 & 0 & 0 \end{bmatrix} = i\epsilon_{123}T_3$$

$$[T_2, T_3] = \frac{1}{4} \left(\begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 0 & i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = i\epsilon_{231}T_1$$

$$[T_1, T_3] = \frac{1}{4} \left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = i\epsilon_{132}T_2$$

□

I don't quite understand what the question means. If it means that to reduce the Gell Mann representation of Lie algebra $su(3)$ restricted on the $su(2)$ sub-algebra, then it is obvious since

$$T_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \sigma_1 \otimes 0, T_2 = \sigma_2 \otimes 0, T_3 = \sigma_3 \otimes 0$$

Then the 3d representation can be reduced to $2 \oplus 1$ representation, where the irreducible representation on the 2d subspace are Pauli matrices, and on 1d subspace are zeros.

Same as above, consider the restriction of the adjoint representation (which is a 8d representation) on the $su(2)$ sub-algebra.

$$ad_{T_1} = i|T_3\rangle\langle T_2| + i\frac{1}{2}|T_7\rangle\langle T_4| - \frac{1}{2}i|T_6\rangle\langle T_5| + h.c.$$

$$ad_{T_2} = -i|T_3\rangle\langle T_1| + \frac{1}{2}i|T_6\rangle\langle T_4| + \frac{1}{2}i|T_7\rangle\langle T_5| + h.c.$$

$$ad_{T_3} = i|T_2\rangle\langle T_1| + \frac{1}{2}i|T_5\rangle\langle T_4| - \frac{1}{2}i|T_7\rangle\langle T_6| + h.c.$$

And we see that by transforming the matrices into Jordan canonical form, the 8d representation can be reduced to

$$(T_1, T_2, T_3) \oplus (T_4 + iT_5, T_6 + iT_7) \oplus (T_4 - iT_5, T_6 - iT_7) \oplus (T_8) = 3 \oplus 2 \oplus 2 \oplus 1$$

Where the representation in 3d subspace is isomorphic to angular momentum algebra, the representations in 2d subspace are isomorphic to the fundamental representation of $su(2)$ algebra, while the representation in 1d subspace are zeros.

PROBLEM 7. Consider the $SU(2)$ subalgebra of $SU(3)$.

- (10 points) Show that λ_2, λ_5 , and λ_7 generate an $SU(2)$ subalgebra of $SU(3)$.

- (20 points) How does the representation generated by the Gell-Mann matrices transform under this subalgebra?

- (20 points) Answer the same question for the adjoint representation of $SU(3)$. That is, How does the adjoint representation of $SU(3)$ transform under this subalgebra?

SOLUTION:

It suffices to prove that $\lambda_1, \lambda_2, \lambda_3$ forms a closed algebra and satisfies the relation

$$[\lambda_a, \lambda_b] = i\epsilon_{abc}\lambda_c$$

By direct calculation we see that

$$[\lambda_5, \lambda_2] = \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = -i\lambda_7$$

$$\begin{aligned}
[\lambda_5, \lambda_7] &= \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = i\lambda_2 \\
[\lambda_2, \lambda_7] &= \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} = -i\lambda_5
\end{aligned}$$

□

I really don't understand what does it mean...

$$\lambda_i \rightarrow [\lambda_{2,5,7}, \lambda_i]$$

Same as above, consider the restriction of the adjoint representation (which is a 8d representation) on the $su(2)$ sub-algebra.

$$\begin{aligned}
ad_{\lambda_2} &= 2(-i|\lambda_3\rangle\langle\lambda_1| + \frac{1}{2}i|\lambda_6\rangle\langle\lambda_4| + \frac{1}{2}i|\lambda_7\rangle\langle\lambda_5| + h.c.) \\
ad_{\lambda_5} &= 2(\frac{1}{2}i|\lambda_6\rangle\langle\lambda_1| - i\frac{1}{2}|\lambda_7\rangle\langle\lambda_2| - \frac{\sqrt{3}}{2}i|\lambda_8\rangle\langle\lambda_4| + h.c.) \\
ad_{\lambda_7} &= 2(\frac{1}{2}i|\lambda_4\rangle\langle\lambda_1| + \frac{1}{2}i|\lambda_5\rangle\langle\lambda_2| - \frac{1}{2}i|\lambda_6\rangle\langle\lambda_3| - \frac{3}{2}i|\lambda_6\rangle\langle\lambda_8| + h.c.)
\end{aligned}$$